

C-Tech SBR (Sequencing Batch Reactor)

Intelligent Cyclic Activated Sludge Wastewater Treatment.

C-Tech Operating Cycle

Phase	Typical duration	Main function
Fill and aeration	1.5 hours	Influent reception, BOD removal, nitrification and denitrification
Settling	0.75 hours	Quiescent biological sludge settling
Decanting	0.75 hours	Removal of clarified supernatant
Total cycle	3.0 hours	Complete treatment cycle
Cycles per basin per day	8 cycles/day	24/324/324/3
Aeration per basin per day	12 hours/day	1.5x81.5 \times 81.5x8

Biological Selector Data

Parameter	Example value
Selector configuration	Six compartments per basin
Selector operating condition	Anoxic/anaerobic mixing
RAS recycle ratio	25% of basin feed flow
Feed flow to each basin	1,041.67 m ³ /h
RAS flow	Approximately 260.42 m ³ /h
RAS pump capacity	Approximately 265 m ³ /h
Combined selector flow	Approximately 1,306.67 m ³ /h
Selector HRT	55 minutes
Selector volume	Approximately 1,282.5 m ³ per basin
Selector width	Approximately 5 m
Selector side-water depth	Approximately 5.7 m

Parameter	Example value
The selector receives raw sewage together with recycled activated sludge. High BOD and low-DO conditions favor floc-forming and phosphorus-accumulating organisms while limiting filamentous growth.	